# Evaluation runs: commit (s3-commit)

generated 2026-11-05 from brain-data 0799b9755 2026-11-05 11:44:26 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): (the single pair): expected conforms. Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair single, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,0 | 4 | 4 | conforms | 0.0 | 4 | 1 h, 250 GB | 4 | 10 | brain03, 2026-11-02 |
| 2,1 | 4 | 4 | conforms | 0.0 | 4 | 1 h, 250 GB | 4 | 10 | brain03, 2026-11-02 |
| 2,2 | 4 | 4 | conforms | 0.0 | 2 | 1 h, 250 GB | 4 | 10 | brain03, 2026-11-02 |
| 4,0 | 16 | 16 | conforms | 0.0 | 2 | 1 h, 250 GB | 16 | 136 | brain03, 2026-11-02 |
| 4,2 | 16 | 16 | conforms | 0.0 | 4 | 1 h, 250 GB | 16 | 136 | brain03, 2026-11-02 |
| 4,4 | 16 | 16 | conforms | 0.0 | 4 | 1 h, 250 GB | 16 | 136 | brain03, 2026-11-02 |
| 8,0 | 256 | 256 | conforms | 1.5 | 6 | 1 h, 250 GB | 256 | 32896 | brain03, 2026-11-02 |
| 8,4 | 256 | 256 | conforms | 1.5 | 6 | 1 h, 250 GB | 256 | 32896 | brain03, 2026-11-02 |
| 8,8 | 256 | 256 | conforms | 1.1 | 6 | 1 h, 250 GB | 256 | 32896 | brain03, 2026-11-02 |
| 12,0 | 4096 | 4096 | conforms | 604.2 | 160 | 1 h, 250 GB | 4096 | 8390656 | brain03, 2026-11-02 |
| 12,6 | 4096 | 4096 | conforms | 678.9 | 82 | 1 h, 250 GB | 4096 | 8390656 | brain03, 2026-11-02 |
| 12,12 | 4096 | 4096 | conforms | 499.6 | 49 | 1 h, 250 GB | 4096 | 8390656 | brain03, 2026-11-02 |
| 16,0 | 65536 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 2 4 fifo, 2 6 fifo, 2 8 fifo, 2 10 fifo, 2 12 fifo, 2 16 fifo, 2 20 fifo, 3 0 fifo, 3 1 fifo, 3 2 fifo, 3 4 fifo, 3 6 fifo, 3 8 fifo, 3 10 fifo, 3 12 fifo, 3 16 fifo, 3 20 fifo, 4 1 fifo, 4 6 fifo, 4 8 fifo, 4 10 fifo, 4 12 fifo, 4 16 fifo, 4 20 fifo, … (160 instances in all).

skipped: a smaller size had reached its limit: 16 1 fifo, 16 2 fifo, 16 8 fifo, 16 16 fifo, 17 0 fifo, 17 1 fifo, 17 2 fifo, 17 8 fifo, 17 16 fifo, 18 0 fifo, 18 1 fifo, 18 2 fifo, 18 8 fifo, 18 16 fifo, 19 0 fifo, 19 1 fifo, 19 2 fifo, 19 8 fifo, 19 16 fifo, 20 0 fifo, 20 1 fifo, 20 2 fifo, 20 8 fifo, 20 10 fifo, … (47 instances in all).

## stateful

### stateful: pair single, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,0 | 4 | 4 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,1 | 4 | 4 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,2 | 4 | 4 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,0 | 16 | 16 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 16 | 16 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | 16 | 16 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,0 | 256 | 256 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,4 | 256 | 256 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,8 | 256 | 256 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 12,0 | 4096 | 4096 | conforms | 1.7 | 32 | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 12,6 | 4096 | 4096 | conforms | 1.7 | 36 | 1 h, 250 GB | brain02, 2026-11-02 |
| 12,12 | 4096 | 4096 | conforms | 1.9 | 34 | 48 h, 250 GB | brain04, 2026-11-03 |
| 16,0 | 65536 | 65536 | conforms | 1007.8 | 643 | 1 h, 250 GB | brain02, 2026-11-02 |
| 16,8 | 65536 | 65536 | conforms | 1013.1 | 643 | 1 h, 250 GB | brain02, 2026-11-02 |
| 16,16 | 65536 | 65536 | conforms | 958.0 | 643 | 48 h, 250 GB | brain04, 2026-11-03 |
| 20,0 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): 2 4 fifo, 2 6 fifo, 2 8 fifo, 2 10 fifo, 2 12 fifo, 2 16 fifo, 2 20 fifo, 3 0 fifo, 3 1 fifo, 3 2 fifo, 3 4 fifo, 3 6 fifo, 3 8 fifo, 3 10 fifo, 3 12 fifo, 3 16 fifo, 3 20 fifo, 4 1 fifo, 4 6 fifo, 4 8 fifo, 4 10 fifo, 4 12 fifo, 4 16 fifo, 4 20 fifo, … (185 instances in all).

skipped: a smaller size had reached its limit: 20 1 fifo, 20 2 fifo, 20 10 fifo, 20 20 fifo, 21 0 fifo, 21 1 fifo, 21 2 fifo, 21 10 fifo, 21 20 fifo, 22 0 fifo, 22 1 fifo, 22 2 fifo, 22 10 fifo, 22 20 fifo, 23 0 fifo, 23 1 fifo, 23 2 fifo, 23 10 fifo, 23 20 fifo.

# offline

## offline: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,0 | 4 | 4 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,1 | 4 | 4 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,2 | 4 | 4 | conforms | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,0 | 16 | 16 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,2 | 16 | 16 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,4 | 16 | 16 | conforms | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,0 | 256 | 256 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,4 | 256 | 256 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,8 | 256 | 256 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 12,0 | 4096 | 4096 | conforms | 1.6 | 74 | 10 min, 16 GB | 0.4 | 0.4 | 1.1 | in-memory | brain01, 2026-11-02 |
| 12,6 | 4096 | 4096 | conforms | 1.5 | 74 | 10 min, 16 GB | 0.4 | 0.4 | 1.1 | in-memory | brain01, 2026-11-02 |
| 12,12 | 4096 | 4096 | conforms | 0.6 | 8 | 48 h, 250 GB | 0.3 | 0.4 | 0.2 | merge | brain01, 2026-11-03 |
| 16,0 | 65536 | 65536 | conforms | 12.7 | 78 | 1 h, 250 GB | 9.0 | 9.1 | 3.5 | merge | brain01, 2026-11-03 |
| 16,8 | 65536 | 65536 | conforms | 12.7 | 78 | 1 h, 250 GB | 8.9 | 9.0 | 3.7 | merge | brain01, 2026-11-03 |
| 16,16 | 65536 | 65536 | conforms | 12.4 | 78 | 1 h, 250 GB | 5.4 | 8.9 | 3.5 | merge | brain01, 2026-11-03 |
| 20,0 | 1048576 | 1048576 | conforms | 274.3 | 1190 | 1 h, 250 GB | 206.9 | 206.2 | 67.5 | merge | brain01, 2026-11-03 |
| 20,10 | 1048576 | 1048576 | conforms | 272.6 | 1189 | 1 h, 250 GB | 205.3 | 204.7 | 67.3 | merge | brain01, 2026-11-03 |
| 20,20 | 1048576 | 1048576 | conforms | 274.2 | 1191 | 1 h, 250 GB | 111.5 | 206.9 | 67.3 | merge | brain01, 2026-11-03 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 2 4 fifo, 2 6 fifo, 2 8 fifo, 2 10 fifo, 2 12 fifo, 2 16 fifo, 2 20 fifo, 3 0 fifo, 3 1 fifo, 3 2 fifo, 3 4 fifo, 3 6 fifo, 3 8 fifo, 3 10 fifo, 3 12 fifo, 3 16 fifo, 3 20 fifo, 4 1 fifo, 4 6 fifo, 4 8 fifo, 4 10 fifo, 4 12 fifo, 4 16 fifo, 4 20 fifo, … (202 instances in all).